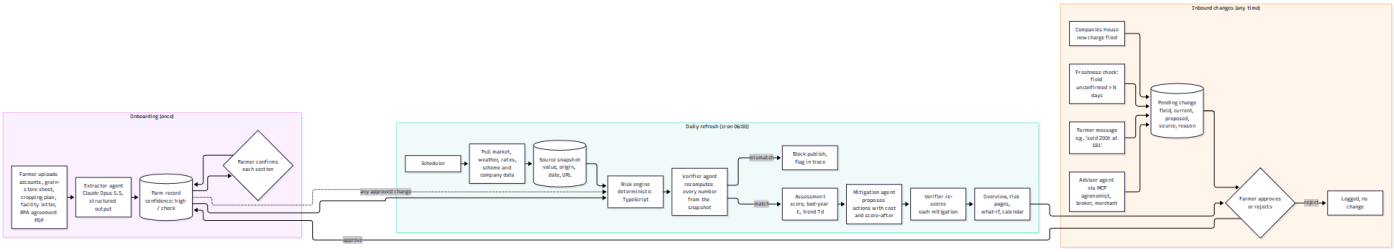
Headwind: agent architecture and end-to-end flow

Status: draft for the AgentVerse build, 7-8 Nov 2026. Fictional farm, real data sources.

Headwind turns a UK arable farm's position (tonnes unsold, nitrogen unbought, debt unfixed, scheme agreements expiring) plus live market, weather and policy data into one number per risk: the £ hit in a 1-in-20 bad year, as a share of a typical year's profit. It then proposes mitigations, and nothing about the farm's record changes without the farmer approving it.

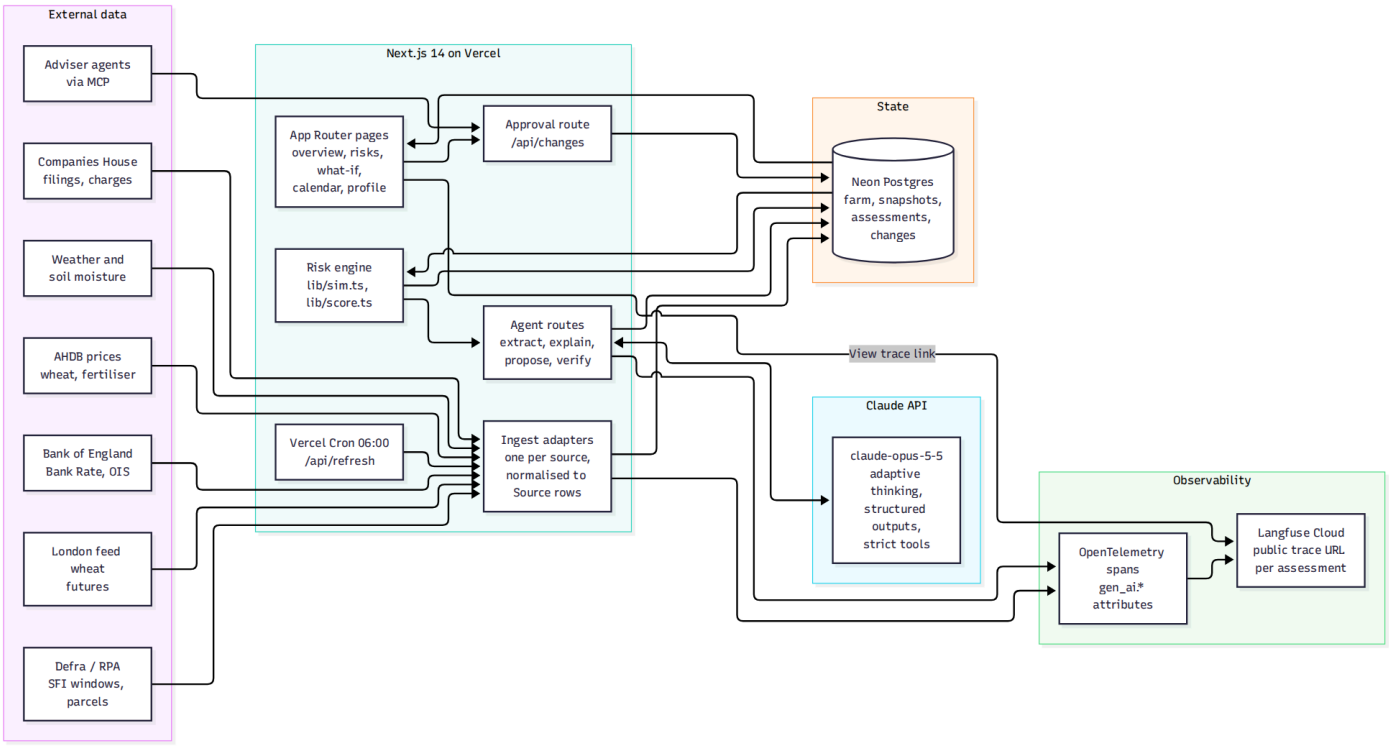
1. End-to-end process



Rules the flow enforces:

- Every figure on screen traces to a row in the source snapshot with origin and date. The verifier refuses to publish an assessment whose numbers it cannot reproduce.
- Only the farmer writes to the farm record. Agents and external feeds can only create pending changes.
- The risk maths is plain TypeScript (`lib/sim.ts`, `lib/score.ts`), not model output. Models extract, explain, propose and verify. They never compute the score.

2. System architecture



3. Data sources

Verified 7 Nov 2026 by fetching each endpoint. "No API" means the realistic path is a spreadsheet download or an HTML scrape on the cron, cached as a JSON snapshot in the repo so the demo survives a blob move.

3.1 Market prices

Need	Source	Endpoint	Format	Auth	Frequency	Licence
Delivered feed wheat, spot and forward by region	AHDB UK delivered prices	`projectblue.blob.core.windows.net/media/Default/MI Reports/D&A Arable/Daily and Weekly Price Reports/UK Delivered Cereals Oilseed Prices.xlsx`	XLSX, latest week only	none	Weekly, Friday	AHDB terms, internal business use
Ex-farm wheat history 2014 to now, monthly by region	AHDB Corn Returns	same blob folder, `Corn Returns prices month_year.xlsx`	XLSX, 2,158 rows	none	Weekly, Monday	AHDB terms
ICE UK feed wheat futures strip	AHDB markets page	`https://ahdb.org.uk/cereals-oilseeds-markets` table "UK (ICE) Feed Wheat £/t"	HTML, server-rendered	none	Daily, weekdays	AHDB terms
Ammonium nitrate £/t, weekly since Mar 2026 plus 438 monthly rows of history	AHDB GB fertiliser price series	`projectblue.blob.core.windows.net/media/Default/Market Intelligence/GBFertiliserPriceSeries.xlsx`	XLSX	none	Weekly, Friday	AHDB terms
Licence-clean index for nitrogen and cereals, 2014 to now	Defra Agricultural Price Index	resolve via `https://www.gov.uk/api/content/government/statistics/agricultural-price-indices` then `details.attachments[].url`	CSV, 2020 = 100	none	Monthly	OGL v3

AHDB has no API. Hectare Trading market insights sit behind a login with no public API. ICE's own delayed-quote JSON and Barchart pages were not verified and are likely bot-blocked.

3.2 Weather, soil and yield

Need	Source	Endpoint	Format	Auth	Limits	Licence
Forecast, rain, ET0, soil moisture at four depths	Open-Meteo forecast	`https://api.open-meteo.com/v1/forecast?latitude=&longitude=&hourly=soil_moisture_0_to_1cm,soil_moisture_27_to_81cm,precipitation,et0_fao_evapotranspiration`	JSON	none	10k calls/day free, non-commercial	CC BY 4.0
Seasonal outlook to 9 months, ensemble members	Open-Meteo seasonal	`https://seasonal-api.open-meteo.com/v1/seasonal?...&forecast_days=45`	JSON	none	as above	CC BY 4.0
10-year rain and ET0 for yield-stress proxy	Open-Meteo archive (ERA5)	`https://archive-api.open-meteo.com/v1/archive?...&start_date=2016-01-01`	JSON	none	5-day lag	CC BY 4.0
Met Office branding on the forecast	Met Office Weather DataHub	`https://data.hub.api.metoffice.gov.uk/sitespecific/v0`, header `apikey` (header name unverified)	JSON	free key	360 calls/day	Met Office terms
Regional wheat yield 1999 to 2026, 5-year average	Defra cereal and OSR production	`https://assets.publishing.service.gov.uk/media/6ac66b15f1ba855cc196a8c8/UK-cops-webseries_261008.ods` sheet `Regional_wheat`	ODS	none	Annual	OGL v3

Open-Meteo is the only true live JSON API in the stack. Copernicus CDS is queued and slow, so skip it.

3.3 Land, schemes, finance, company

Need	Source	Endpoint	Format	Auth	Limits	Licence
Parcel locations near the farm	RPA Parcel Points (England)	`https://environment.data.gov.uk/spatialdata/rpa-parcel-points-england/ogc/features/v1/collections/.../items?bbox=`	GeoJSON, centroids only	none	annual	OGL v3
Live SFI actions by parcel	SFI Scheme Actions (England)	dataset `environment.data.gov.uk/dataset/58cc85ab-...` OGC Features	GeoJSON	none	periodic	OGL v3
Scheme action payment rates	gov.uk content API	`https://www.gov.uk/api/search.json?filter_content_store_document_type=farming_grant&count=203` then `/api/content/find-funding-for-land-or-farms/<slug>`	JSON, rates inside body HTML	none	none	OGL v3
SFI26 windows and caps	Defra blog and news	hard-code: window 1 June 2026, window 2 opened 22 Sep 2026 and closed within a day, £100k/yr cap, early renewal for agreements expiring by 28 Feb 2027	HTML	none	static	OGL
Bank Rate and SONIA, daily	Bank of England IADB	`https://www.bankofengland.co.uk/boeapps/database/_iadb-fromshowcolumns.asp?csv.x=yes&Datefrom=01/Jan/2016&Dateto=now&SeriesCodes=IUDBEDR,IUDSOIA&CSVF=TN&UsingCodes=Y&VPD=Y&VFD=N`	CSV	none	daily	BoE terms
OIS curve	Bank of England yield curves	`oisddata.zip`, 12 MB	ZIP of XLSX	none	daily	BoE terms
Charges, filings, accounts	Companies House Public Data API	`https://api.company-information.service.gov.uk/company/{number}/filing-history`	JSON	free key as Basic auth username	600 requests per 5 min	CH terms

RPA does not publish parcel polygons openly, so draw mock boundaries around the points. Rural Payments has no third-party API. Most arable farms are partnerships with no Companies House record, so the demo farm stays a Ltd. Farm software (Yagro, Gatekeeper, Omnia) exposes no public API, so adviser data arrives as CSV upload or via a mocked MCP server.

3.4 Minimal set for the 33-hour build

1. Open-Meteo forecast, archive and seasonal. Live, JSON, no key.
2. Bank of England IADB CSV for Bank Rate and SONIA. One fetch.
3. The three AHDB spreadsheets, parsed with the `xlsx` package on the cron and committed as JSON snapshots.
4. AHDB markets HTML table for the ICE forward strip, or the hard-coded strip from 7 Nov: Nov-26 189.00, Jan-27 192.25, Mar-27 195.50, May-27 198.75, Jul-27 202.00.
5. Defra price-index CSV as the OGL-licensed volatility series shown in the pitch.
6. gov.uk content API plus hard-coded SFI26 dates for the Schemes lane.
7. RPA parcel points for the profile map.
8. Companies House only because the demo farm is a Ltd.
9. Defra regional wheat yield ODS, parsed once and committed.

4. Agent design

Four model roles, one engine. All calls use `claude-opus-5-5` with adaptive thinking (the default), through `@anthropic-ai/sdk`.

Role	Trigger	Input	Output	How
Extractor	Onboarding upload	PDF or XLSX as a `document` block	`OnboardingField[]` with value, source, confidence	`client.messages.parse()` with a Zod schema; `confidence: "check"` for anything inferred rather than read
Explainer	After each assessment	Assessment plus the source snapshot rows	`reasoning[]` and `summary` for a risk page	Structured output. The prompt forbids any number not present in the snapshot; the verifier checks this
Proposer	After each assessment	Risk, farm record, available instruments and quotes	`Mitigation[]` with cost, units removed, status "suggested"	Strict tools: `quote_min_price`, `quote_forward`, `quote_fixed_rate`, each a thin adapter over a source or a mock
Verifier	Before publish	Assessment, mitigations, snapshot, the engine functions as tools	`{ ok, mismatches[] }`	Tool runner with `recompute_bad_year` and `recompute_score` tools bound to `lib/sim.ts` and `lib/score.ts`. Any number the verifier cannot reproduce within rounding blocks publish

Why the engine stays deterministic: the judges want reasoning a professional can check. A 1-in-20 figure that comes from a z-score times a standard deviation times exposed tonnes is checkable on a napkin. A figure that comes from a model is not. The models only read documents, write prose, propose instruments and audit.

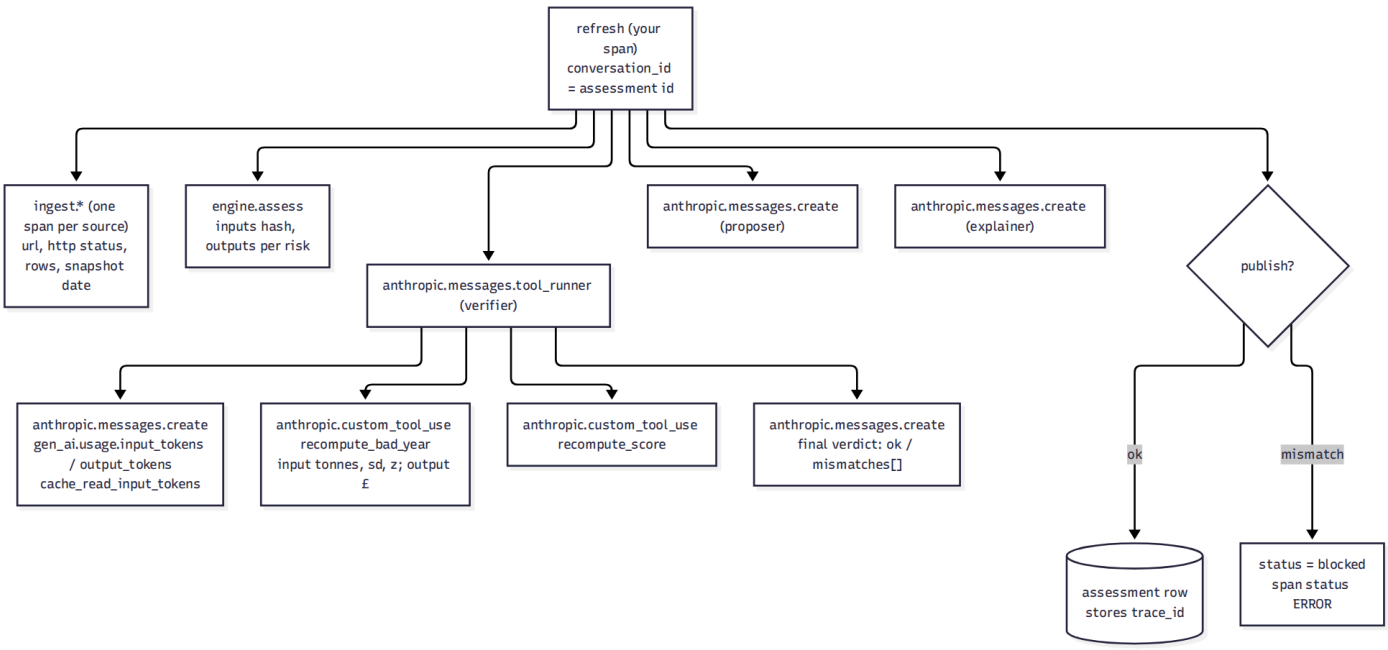
Human in the loop: every inbound signal (freshness timer, farmer message, adviser MCP, Companies House) becomes a `PendingChange`. The farm record only changes through the approval route. Agents never hold a write tool to the farm record.

Prompt caching: system prompt and tool list first and frozen, then the snapshot with a `cache_control` breakpoint, then the per-risk question. Verify with `usage.cache_read_input_tokens` on the second call of each refresh.

5. Observability

Goal on stage: open any risk page, click "View trace", and show the judges the exact prompts, tool calls, recomputed numbers, token cost and latency behind the figure on screen.

5.1 Trace tree per refresh



5.2 Mechanics

The Anthropic TypeScript SDK emits OpenTelemetry spans itself (beta), with `gen_ai.*` attributes, so any OTLP backend reads them without mapping. Two settings matter for the demo:

- `traces.contentMode: "content"` so prompts and responses are in the span. Default is metadata only, which would make the trace useless to a judge.
- `conversationId` set to the assessment id on every call in one refresh, so the backend groups the verifier, proposer and explainer under one conversation.

The tracer provider is registered once in `instrumentation.ts` (final version in 5.4).

```
// lib/claude.ts
import Anthropic from "@anthropic-ai/sdk";
export const client = new Anthropic({
  openTelemetry: { traces: { contentMode: "content" } },
});

// per call inside the refresh
await client.messages.create(params, { openTelemetry: { conversationId: assessmentId } });
```

Wrap the whole refresh in one span from `@opentelemetry/api` (`tracer.startActiveSpan("refresh", ...)`) so the SDK spans nest under it, and record `span.spanContext().traceId` on the assessment row. The risk page links to `${TRACE_BASE_URL}/${traceId}`.

Vercel serverless note: flush the span processor before the route returns, or spans are lost when the function freezes.

5.3 What the verifier span must contain

For the "checkable by a professional" criterion, each `recompute_*` tool span carries the inputs and output as attributes, not just in the prompt text. A judge can read: tonnes exposed 2,898, price sd £20/t, z 1.645, bad-year £95.3k, engine said £96k, within rounding. That is the whole observability story in one span.

5.4 Backend choice

Verified 7 Nov 2026 unless marked. Setup time assumes Next.js on Vercel with an `instrumentation.ts`.

Platform	Free tier	Setup	Ingests OTel GenAI spans	Tool calls, I/O, cost, latency	Evals	Public trace link
Langfuse Cloud (EU/US, MIT self-host)	Hobby: no card, 50k units/month, 2 users	15 min, one span processor and two env vars	Yes, native OTLP; Anthropic lists it as a built-in API-tracing destination	Yes	LLM-as-judge, annotation queue, scores API	Yes, one call per trace
LangSmith	Developer: 1 seat, 5k traces/month then $2.50 per 1k	15 min	Yes, OTLP endpoint	Yes	Datasets, evaluators	Share links (unverified)
Arize Phoenix	OSS free; cloud 25k spans/month, 15-day retention (secondary source)	20 min	Yes, OpenInference plus GenAI	Yes	Evals library	Unverified
Braintrust	Starter: 1 GB data, 14-day retention (secondary source); Pro $249/month	15 min	Yes	Yes	Strongest, CI gating	Unverified
Helicone	10k requests/month, 7-day retention	5 min, base-URL proxy	Proxy model, no tool-execution spans	Cost and latency only	Limited	Unverified
Sentry AI Insights	Developer plan, limits unverified	15 min if Sentry already present	Yes	Yes	None	Org-internal only
Vercel AI Gateway observability	Included	None if routed via Gateway	Not a trace store	Requests, usage, cost, no tool tree	None	No

Decision: **Langfuse Cloud EU** as the only trace store. It is the one option verified to give a public link per trace, which is what lets a judge open the exact run on a phone. It takes both span sources (the Anthropic SDK's own spans and the app's spans) through one tracer provider, and the Anthropic API can additionally be pointed at it server-side by an org admin for model-request spans that nest under ours.

Final `instrumentation.ts`, replacing the generic exporter in 5.2:

```
import { NodeSDK } from "@opentelemetry/sdk-node";
import { LangfuseSpanProcessor } from "@langfuse/otel";

export async function register() {
  if (process.env.NEXT_RUNTIME !== "nodejs") return;
  const sdk = new NodeSDK({ spanProcessors: [new LangfuseSpanProcessor()] });
  sdk.start(); // global provider; @anthropic-ai/sdk spans land here with no further wiring
}
```

```
LANGFUSE_PUBLIC_KEY=pk-lf-...
LANGFUSE_SECRET_KEY=sk-lf-...
LANGFUSE_BASE_URL=https://cloud.langfuse.com
ANTHROPIC_API_KEY=sk-ant-...
ANTHROPIC_OPEN_TELEMETRY_TRACES_CONTENT_MODE=content
CRON_SECRET=<random>
```

Make the trace public from the refresh span with `startObservation(...).setTraceAsPublic()` from `@langfuse/tracing`, and store its trace id on the assessment row. Run `LangfuseSpanProcessor.forceFlush()` before the cron route returns.

Packages: `@anthropic-ai/sdk @opentelemetry/sdk-node @langfuse/otel @langfuse/tracing zod`. The Vercel AI SDK is not needed; the Anthropic SDK's tool runner and `messages.parse()` cover the loop and structured output.

5.5 Runtime notes

- Vercel Cron on Hobby fires once a day with up to 59 minutes of drift, so `0 6 * * *` lands between 06:00 and 06:59. Fallback: a GitHub Actions schedule hitting `/api/refresh` with the `CRON_SECRET` bearer token.
- Storage: Neon Postgres (free 512 MB, Vercel Marketplace, does not pause). Supabase's free project pauses after 7 idle days, which is a demo-day risk. Tables: `farm`, `source_snapshot`, `assessment` (with `trace_id`), `pending_change`.
- Human in the loop needs no durable-execution library. The proposer writes `pending_change` rows; the approval route flips `status`; the next refresh reads approved rows. The agent run ends after proposing.

6. Hackathon scope

Build order, each step demoable on its own:

1. Cron route that pulls Open-Meteo and the BoE CSV, writes a snapshot, re-runs the engine. Overview page shows a live `updatedAt`.
2. AHDB spreadsheets parsed to JSON, replacing the hard-coded £178/t and £504/t. Sources panel shows real dates.
3. Verifier agent with trace link on every risk page. This is the observability demo.
4. Pending-change approval wired to the farm record, with one real inbound source (Companies House filing-history poll).
5. Extractor on one uploaded PDF, if time allows.

Out of scope: yield-index insurance quotes (no UK product to quote), live option premiums from merchants (no API), parcel polygons (not open data).